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question

Guideline

There are 5 questions in the exam. For Question 1 - 4, Write your answer in a separate paper/file, take picture of it (Use camscanner) and create a pdf file containing all answer. For each question there is a submit button. You can simply click on submit button. You need to submit the merged PDF at the end of the question (There is an option for uploading file)

For question number 5, You MUST submit in buX.

While drawing the diagrams you can draw it by hand or use any tool (Optional). Drawing in tools might take longer time. So you are advised to avoid tools.

This site can be used to convert images to pdf: <https://smallpdf.com/>

Also, it would be good if they can compress the files where compression rate is 90%+: <https://www.wecompress.com/en/>

You are given only one attempt to submit the answers. So before pressing Submit button, make sure that is your final answer.

Total duration is 140 minutes. (110 minutes for exam , 30 minutes for file uploading) It is your choice how to utilize the time



Contact us if you failed to complete the exam in time.

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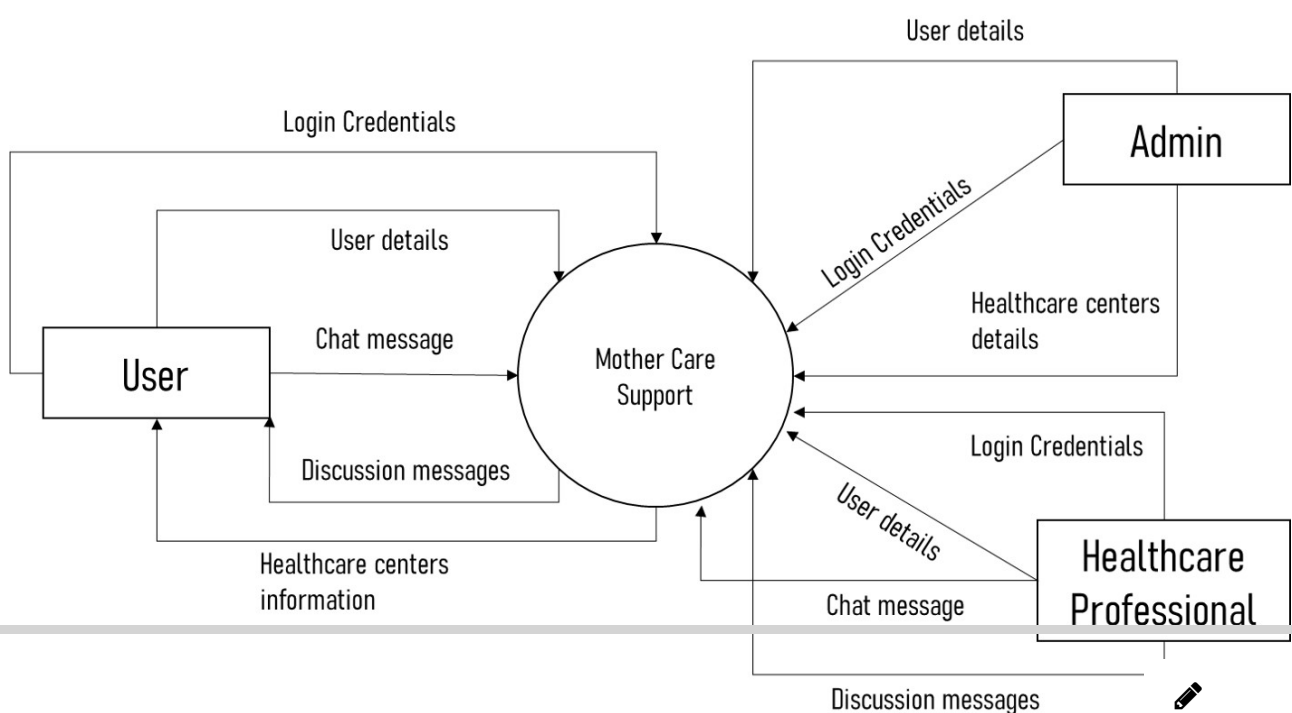
A viva session might be conducted after the exam.

1. Draw an Entity Relationship Diagram (ERD) for “University Management System”

where primary and composite key are required. Your mark is distributed according to your analyzing capability, usage of different relationships, properties, and ease of visual representation where entire database along with all entities and primary/composite key are required.

Total marks 10

2.



BRAC is designing an Interactive Mobile Application for Mother Care Support for Bangladesh. The context diagram is given above.

1. **Design level-1 DFD** based on the following scenario.

Users, administrators and healthcare professionals can create accounts in the mobile application through Register new user process by providing their user details. All user details are stored in user info datastore. Any type of user may log in to the system using the login process by the login credentials. This process can send user details from user info datastore and receive an acknowledgement. User and healthcare professionals can interact with other registered users through main modules namely Live Chat and Discussion forum. These two processes store all chat messages and discussion messages to chat history and discussion history datastore. Admin send healthcare center details into healthcare centers directory process. This process send healthcare centers information to the users.

2. Now decompose healthcare centers directory process and **design level-2 diagram** for it.

Health center details goes to processing healthcare center sub process which send health center details data to health center database and generate healthcare center information process. This process also take health center information from health center database and send healthcare center information to the user. Make sure it is *Balanced*. Also mention if you find any *error* in the scenario.

Total marks 10

3. Medi4all is a healthcare program that generates patients' visits and prescriptions based on conversation recording. Create visit module starts by calling upon the Generate conversation record module to prepare a doctor-patient conversation record. It is then sent to a Parse conversation module which returns a text description of doctor-patient conversation to Create visit module. In order to generate text, Parse conversation module first sends raw audio data to Enhance audio module and receives a noise-reduced modified version. This is then passed onto Convert audio-to-text module which returns the raw text data. After receiving this text data, Create visit module receives a formatting flag from Resolve formatting module and based on this decides whether to use Create regular format or Create med-all format module. Create regular format returns a simply formatted text file using a generic formatter, whereas Create med-all format returns a specially formatted file using standard healthcare templates. This text file is passed onto the Extract prescription module. First it receives the extracted diagnoses, symptoms and prescription information from Parse medical components module. Next for each diagnosis, Create :  stion module which is an on-page connector is called to create a suggestion entr then

module which is an on-page connector is called to create a suggestion entry. Then

Update prediction module which is an off-page connector is called to update the prediction model for given suggestion entry. Finally Extract prescription module returns the complete prescription to Create visit module. Create visit module uses Print item library module to print the given prescription.

Design a structure chart based on the above information. (8 marks)

4.

Hogwarts School of Witchcraft and Wizardry is a British wizarding boarding school teaching the magical arts, located in Scotland inside Hogwarts Castle. You are assigned the responsibility to **design a Windows Navigation diagram** for "Hogwarts School of Witchcrafts and Wizardry". The requirements you need to follow are listed below.

1. We need to ensure that the only people who can see the website are wizards. (A little racism but it is necessary for the secrecy of the wizard world).
2. User can search by i) Student name, ii) House Name, iii) Teacher's name iv) Course name
3. While displaying the result for student name search, it will show
 - i) the name of the student
 - ii) the house which he belongs to.
 - iii) courses the student have taken for current semester
4. House Name search will results into
 - i) List of the students in that house.
 - ii) Professor who is appointed as the head of the house.
5. Teachers name search will result into
 - i) The courses the teacher is taking for this current semester
 - ii) house name if the teacher is appointed to be a head of the house.

What you have to keep in the mind:

You need to make sure that every page is interlinked. For example, if someone searches "Harry potter", it will take him to the page containing information about harry potter. There in the place of the house name "Gryffindor" will be written. Now if the user clicks on "Gryffindor" it will take the searcher to the house name page and so on.

Total marks 7

5

5.0/5.0 points (graded)



Database volume estimation - 5 marks

Consider a RDBMS team is working to store the records of a newly established school in your area. They have identified the id (12), name (62), mobile (65), guardian's email (39) and address (62) as the fields of records. Considering an overhead of 15% on data volume with initial 514 records and a growth rate of 986 records per month, calculate the expected volume of the database table in 6 years.

The values enclosed by the first bracket indicated the sizes of corresponding fields

No need to roundup or precision your answer. Just put the actual value.



You have used 1 of 2 attempts

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